

Kode migration

Ini isi method up() untuk ke-19 migration, dalam urutan dijalankan. Tiap file adalah anonymous class Laravel biasa; method down() cukup berisi dropIfExists untuk tabel yang dibuat, kecuali disebut lain. Karena nama tabel tidak jamak, setiap model perlu mengisi properti \$table dan setiap constrained() menyebut nama tabelnya.

Users dan tingkat (migration 1 sampai 3)

Tiga migration ini menyiapkan users untuk soft delete dan menghubungkannya ke tingkat. Migration 1 ditulis tanpa melihat tabel users di repo, jadi tiap perubahan dijaga dengan pengecekan kolom.

1. alter_users_table

```
Schema::table('users', function (Blueprint $table) {
    if (! Schema::hasColumn('users', 'is_active')) {
        $table->boolean('is_active')->default(true);
    }
    if (! Schema::hasColumn('users', 'deleted_at')) {
        $table->softDeletesTz();
    }
    foreach (['role', 'kelas', 'sekolah'] as $kolom) {
        if (Schema::hasColumn('users', $kolom)) {
            $table->dropColumn($kolom);
        }
    }
    $table->dropUnique('users_email_unique');
});

DB::statement(
    'CREATE UNIQUE INDEX users_email_aktif_unique
    ON users (email) WHERE deleted_at IS NULL'
);

Schema::dropIfExists('sessions');
```

Method down() mengembalikan unique email biasa dan menghapus index parsial. Kolom yang dibuang tidak dikembalikan.

2.create_tingkat_seleksi_table

```
Schema::create('tingkat_seleksi', function (Blueprint $table) {
    $table->id();
    $table->string('nama_tingkat', 100);
    $table->text('deskripsi')->nullable();
    $table->unsignedSmallInteger('urutan')->unique();
    $table->timestampsTz();
});
```

3.add_tingkat_aktif_to_users

```
Schema::table('users', function (Blueprint $table) {
    $table->foreignId('tingkat_aktif_id')
        ->nullable()
        ->constrained('tingkat_seleksi')
        ->onDelete();
});
```

Struktur dan bank soal (migration 4 sampai 11)

Delapan migration ini membuat tabel yang diisi admin. Check constraint ditambahkan dengan DB::statement setelah tabel dibuat.

4.create_kompetensi_table

```
Schema::create('kompetensi', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->string('nama_kompetensi', 150);
    $table->text('deskripsi')->nullable();
    $table->timestampsTz();

    $table->unique(['tingkat_id', 'nama_kompetensi']);
});
```

5. create_materi_table

```
Schema::create('materi', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->foreignId('kompetensi_id')->constrained('kompetensi')->restrictOnDelete();
    $table->unsignedSmallInteger('urutan');
    $table->string('judul', 200);
    $table->text('deskripsi')->nullable();
    $table->longText('isi_materi');
    $table->string('file_materi')->nullable();
    $table->timestampsTz();

    $table->unique(['tingkat_id', 'urutan']);
});
```

6. create_konteks_soal_table

```
Schema::create('konteks_soal', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->string('judul', 200);
    $table->text('isi_konteks');
    $table->string('gambar')->nullable();
    $table->timestampsTz();
});
```

7. create_soal_table

```
Schema::create('soal', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->foreignId('materi_id')->constrained('materi')->restrictOnDelete();
    $table->foreignId('konteks_id')->nullable()->constrained('konteks_soal')->restrictOnDelete();
    $table->string('level', 10);
    $table->string('peruntukan', 10);
    $table->string('tipe_soal', 20);
    $table->text('pertanyaan');
    $table->jsonb('pilihan_jawaban')->nullable();
});
```

```

$table->string('kunci_jawaban', 255);
$table->timestampsTz();
$table->softDeletesTz();

$table->index(['tingkat_id', 'peruntukan', 'level']);
$table->index(['materi_id', 'peruntukan']);
});

DB::statement("ALTER TABLE soal ADD CONSTRAINT soal_level_check
CHECK (level IN ('mudah', 'sedang', 'sulit'))");
DB::statement("ALTER TABLE soal ADD CONSTRAINT soal_peruntukan_check
CHECK (peruntukan IN ('pretest', 'latihan', 'simulasi'))");
DB::statement("ALTER TABLE soal ADD CONSTRAINT soal_tipe_check
CHECK (tipe_soal IN ('pilihan_ganda', 'isian'))");

```

8.create_pembahasan_table

```

Schema::create('pembahasan', function (Blueprint $table) {
    $table->id();
    $table->foreignId('soal_id')->unique()->constrained('soal')-
>cascadeOnDelete();
    $table->text('isi_pembahasan');
    $table->timestampsTz();
});

```

9.create_aturan_pemetaan_table

```

Schema::create('aturan_pemetaan', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')-
>restrictOnDelete();
    $table->string('parameter', 60);
    $table->string('ketentuan', 255);
    $table->timestampsTz();

    $table->unique(['tingkat_id', 'parameter']);
});

```

10. create_simulasi_table

```
Schema::create('simulasi', function (Blueprint $table) {
    $table->id();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->string('nama_simulasi', 150);
    $table->text('deskripsi')->nullable();
    $table->unsignedSmallInteger('jumlah_soal');
    $table->unsignedSmallInteger('durasi_menit');
    $table->boolean('is_aktif')->default(false);
    $table->timestampsTz();
});
```

11. create_quiz_table

```
Schema::create('quiz', function (Blueprint $table) {
    $table->id();
    $table->foreignId('materi_id')->unique()->constrained('materi')->restrictOnDelete();
    $table->string('nama_quiz', 150);
    $table->text('deskripsi')->nullable();
    $table->unsignedSmallInteger('jumlah_soal');
    $table->timestampsTz();
});

DB::statement('ALTER TABLE quiz ADD CONSTRAINT quiz_jumlah_soal_check
CHECK (jumlah_soal >= 10)');
```

Pre-test, pemetaan, dan progress (migration 12 sampai 15)

Empat migration ini membuat tabel putaran: pre-test beserta jawabannya, hasil pemetaan, materi wajib, dan progress belajar.

12. create_pretest_tables

```
Schema::create('pretest', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('tingkat_id')->constrained('tingkat_seleksi')->restrictOnDelete();
    $table->timestampTz('disubmit_pada')->nullable();
    $table->decimal('nilai', 5, 2)->nullable();
    $table->timestampTz('selesai_pada')->nullable();
});
```

```

        $table->timestampsTz();

        $table->index(['user_id', 'tingkat_id']);
    });

    DB::statement(
        'CREATE UNIQUE INDEX pretest_berjalan_unique
        ON pretest (user_id, tingkat_id) WHERE selesai_pada IS NULL'
    );

    Schema::create('pretest_jawaban', function (Blueprint $table) {
        $table->id();
        $table->foreignId('pretest_id')->constrained('pretest')->
        >cascadeOnDelete();
        $table->foreignId('soal_id')->constrained('soal')->restrictOnDelete();
        $table->unsignedSmallInteger('urutan');
        $table->unsignedTinyInteger('bobot');
        $table->string('jawaban_user', 255)->nullable();
        $table->boolean('status_benar')->nullable();
        $table->timestampsTz();

        $table->unique(['pretest_id', 'soal_id']);
    });

```

Method down() menghapus pretest_jawaban lebih dulu, lalu pretest.

13.create_pemetaan_materi_table

```

    Schema::create('pemetaan_materi', function (Blueprint $table) {
        $table->id();
        $table->foreignId('pretest_id')->constrained('pretest')->
        >cascadeOnDelete();
        $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
        $table->foreignId('materi_id')->constrained('materi')->
        >restrictOnDelete();
        $table->unsignedSmallInteger('jumlah_soal');
        $table->unsignedSmallInteger('jumlah_benar');
        $table->unsignedSmallInteger('poin_didapat');
        $table->unsignedSmallInteger('poin_maksimal');
        $table->decimal('persentase', 5, 2);
        $table->unsignedSmallInteger('peringkat');
        $table->timestampsTz();

        $table->unique(['pretest_id', 'materi_id']);
    });

```

14. create_rekomendasi_materi_table

```
Schema::create('rekomendasi_materi', function (Blueprint $table) {
    $table->id();
    $table->foreignId('pretest_id')->constrained('pretest')->
    >cascadeOnDelete();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('materi_id')->constrained('materi')->
    >restrictOnDelete();
    $table->unsignedSmallInteger('prioritas');
    $table->timestampsTz();

    $table->unique(['pretest_id', 'materi_id']);
});
```

15. create_progress_belajar_table

```
Schema::create('progress_belajar', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('materi_id')->constrained('materi')->
    >restrictOnDelete();
    $table->string('status', 10)->default('belajar');
    $table->unsignedSmallInteger('persentase')->default(0);
    $table->timestampTz('tanggal_selesai')->nullable();
    $table->timestampsTz();

    $table->unique(['user_id', 'materi_id']);
});

DB::statement("ALTER TABLE progress_belajar ADD CONSTRAINT
progress_status_check
CHECK (status IN ('belajar', 'selesai'))");
```

Latihan, simulasi, kenaikan, dan view (migration 16 sampai 19)

Empat migration terakhir membuat tabel pengerjaan latihan dan simulasi, catatan kelulusan, dan view riwayat. View harus dijalankan paling akhir.

16. create_quiz_pengerjaan_tables

```
Schema::create('quiz_pengerjaan', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('quiz_id')->constrained('quiz')->restrictOnDelete();
    $table->timestampTz('disubmit_pada')->nullable();
    $table->decimal('nilai', 5, 2)->nullable();
    $table->timestampTz('selesai_pada')->nullable();
    $table->timestampsTz();

    $table->index(['user_id', 'quiz_id']);
});

Schema::create('quiz_jawaban', function (Blueprint $table) {
    $table->id();
    $table->foreignId('pengerjaan_id')->constrained('quiz_pengerjaan')->
    >cascadeOnDelete();
    $table->foreignId('soal_id')->constrained('soal')->restrictOnDelete();
    $table->unsignedSmallInteger('urutan');
    $table->unsignedTinyInteger('bobot');
    $table->string('jawaban_user', 255)->nullable();
    $table->boolean('status_benar')->nullable();
    $table->timestampsTz();

    $table->unique(['pengerjaan_id', 'soal_id']);
});
```

17. create_hasil_simulasi_tables

```
Schema::create('hasil_simulasi', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('simulasi_id')->constrained('simulasi')->
    >restrictOnDelete();
    $table->foreignId('pretest_id')->constrained('pretest')->
    >cascadeOnDelete();
    $table->timestampTz('mulai_pada');
    $table->timestampTz('batas_pada');
    $table->timestampTz('disubmit_pada')->nullable();
    $table->timestampTz('selesai_pada')->nullable();
    $table->decimal('nilai', 5, 2)->nullable();
    $table->unsignedSmallInteger('jumlah_benar')->nullable();
    $table->unsignedSmallInteger('jumlah_salah')->nullable();
    $table->boolean('lulus')->nullable();
    $table->timestampsTz();
```

```

        $table->index(['user_id', 'pretest_id']);
        $table->index(['disubmit_pada', 'batas_pada']);
    });

    DB::statement(
        'CREATE UNIQUE INDEX hasil_simulasi_berjalan_unique
        ON hasil_simulasi (user_id) WHERE selesai_pada IS NULL'
    );

    Schema::create('hasil_simulasi_jawaban', function (Blueprint $table) {
        $table->id();
        $table->foreignId('hasil_simulasi_id')->constrained('hasil_simulasi')->
        >cascadeOnDelete();
        $table->foreignId('soal_id')->constrained('soal')->restrictOnDelete();
        $table->unsignedSmallInteger('urutan');
        $table->unsignedTinyInteger('bobot');
        $table->string('jawaban_user', 255)->nullable();
        $table->boolean('status_benar')->nullable();
        $table->timestampsTz();

        $table->unique(['hasil_simulasi_id', 'soal_id']);
    });

```

Index (disubmit_pada, batas_pada) dipakai command penutupan otomatis yang berjalan tiap menit.

18.create_kenaikan_tingkat_table

```
Schema::create('kenaikan_tingkat', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->foreignId('tingkat_asal_id')->constrained('tingkat_seleksi')->
>restrictOnDelete();
    $table->foreignId('tingkat_tujuan_id')->nullable()-
>constrained('tingkat_seleksi')->restrictOnDelete();
    $table->string('status', 15);
    $table->text('keterangan')->nullable();
    $table->timestampsTz();

    $table->index(['user_id', 'tingkat_asal_id']);
});

DB::statement("ALTER TABLE kenaikan_tingkat ADD CONSTRAINT
kenaikan_status_check
CHECK (status IN ('lulus', 'tidak_lulus'))");

DB::statement(
    "CREATE UNIQUE INDEX kenaikan_lulus_unique
    ON kenaikan_tingkat (user_id, tingkat_asal_id) WHERE status = 'lulus'"
);
```

19.create_riwayat_hasil_view

```
public function up(): void
{
    DB::statement(<<<'SQL'
        CREATE VIEW riwayat_hasil AS
        SELECT 'pretest' AS jenis_hasil, p.id AS referensi_id, p.user_id,
            p.tingkat_id, p.nilai, p.selesai_pada AS tanggal
        FROM pretest p
        WHERE p.selesai_pada IS NOT NULL
        UNION ALL
        SELECT 'latihan', qp.id, qp.user_id, m.tingkat_id, qp.nilai,
qp.selesai_pada
        FROM quiz_pengerjaan qp
        JOIN quiz q ON q.id = qp.quiz_id
        JOIN materi m ON m.id = q.materi_id
        WHERE qp.selesai_pada IS NOT NULL
        UNION ALL
        SELECT 'simulasi', hs.id, hs.user_id, s.tingkat_id, hs.nilai,
hs.selesai_pada
        FROM hasil_simulasi hs
```

```
        JOIN simulasi s ON s.id = hs.simulasi_id
        WHERE hs.selesai_pada IS NOT NULL
    SQL);
}

public function down(): void
{
    DB::statement('DROP VIEW IF EXISTS riwayat_hasil');
}
```

Kode ini belum dijalankan terhadap database; jalankan `php artisan migrate:fresh --seed` di lingkungan lokal dan perbaiki nama index bawaan bila berbeda dari yang diasumsikan di migration 1.